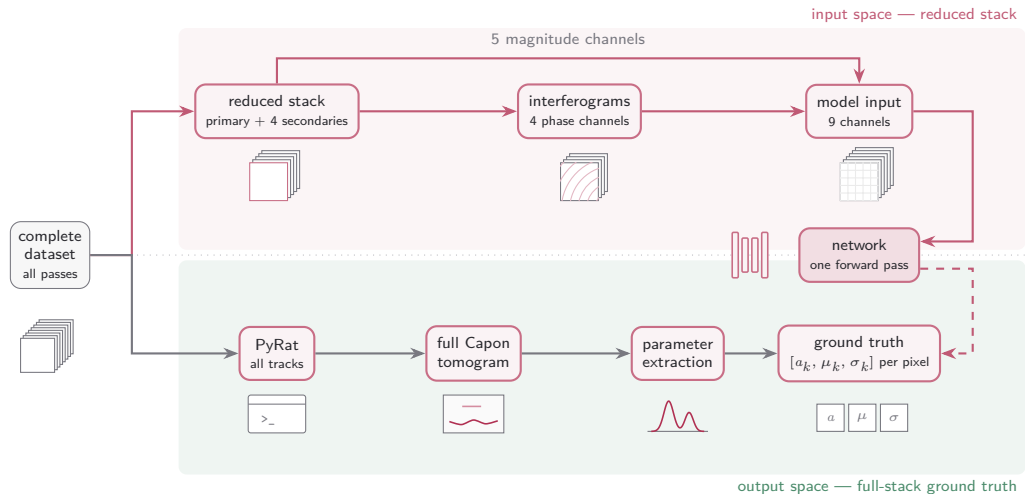


What this project does



A network learns to reconstruct, from the **reduced** stack alone, the per-pixel profile parameters that the full-stack tomogram and classical fit produce.

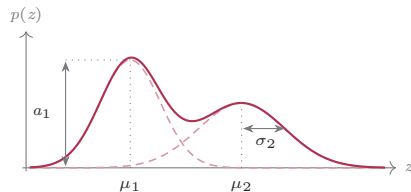
Parameterising the spectrum — and its classical cost

The elevation power spectrum at each pixel is modelled as a K -Gaussian mixture:

$$p(z) = \sum_{k=1}^K a_k \exp\left(-\frac{(z-\mu_k)^2}{2\sigma_k^2}\right)$$

a_k backscattered power of scatterer k
 μ_k its elevation in the cell
 σ_k its vertical spread

Working configuration: $K=2$ — covers 93.9% of active pixels; the $K=5$ extension for the multi-scatterer tail is in development.



Mixture $p(z)$ and its components (dashed) — three numbers per scatterer describe the whole profile.



The classical route restarts an iterative fit at every pixel — the cost the amortised network removes (next).

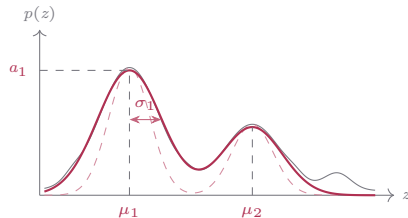
Ground truth: fitted Gaussian parameters

Supervised target: the per-pixel array $[a_k, \mu_k, \sigma_k]_{k=1}^K$ ($K=2$ today, $K=5$ in development), fit to the per-pixel max-normalised Capon spectrum $\tilde{p} = p/p_{\max}$ on the elevation grid ξ_n :

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N (\hat{p}(\xi_n) - \tilde{p}(\xi_n))^2, \quad \hat{p}(\xi) = \sum_k a_k e^{-\frac{(\xi - \mu_k)^2}{2\sigma_k^2}}$$

One shared loss; each parameter group takes its own projected Adam step:

$$\begin{aligned} a_k &\leftarrow [a_k - \eta \text{Adam}(\partial_{a_k} \mathcal{L})]_+ \\ \mu_k &\leftarrow \text{clip}_{[\mu^-, \mu^+]} (\mu_k - \eta \text{Adam}(\partial_{\mu_k} \mathcal{L})) \\ \sigma_k &\leftarrow \text{clip}_{[\sigma^-, \sigma^+]} (\sigma_k - \eta \text{Adam}(\partial_{\sigma_k} \mathcal{L})) \end{aligned}$$

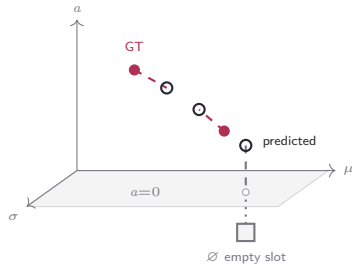


Same pixel: Capon spectrum, **peak initialisation (dashed)**, **converged fit**; the fitted $[a_k, \mu_k, \sigma_k]$ are the label.

- Fitted in normalised space; the winning amplitudes are denormalised back, $a_k \rightarrow p_{\max} a_k$ (μ_k, σ_k already live on the physical axis).
- All pixels batched in one optimisation — not a per-pixel loop (next slide); most pixels activate one or two Gaussians (Act III).

Hungarian matching

- Each Gaussian is a **point in (a, μ, σ) space**: prediction and GT are two point sets; the matcher pairs them by distance and scores the **set**, not the slot layout.



Filled = active GT, open = predicted, $\emptyset$ = empty GT slot (below $a=0$).
Its match (grey) is zero-weighted, so $\hat{\theta}_3$ is free.

$$\mathcal{L} = \min_{\pi \in S_K} \sum_{k=1}^K w_k \left\| \hat{\theta}_{\pi(k)} - \theta_k \right\|^2 \quad \theta_k = (a_k, \mu_k, \sigma_k)$$

$w_k = 1[a_k > \tau]$ — GT active mask: 1 if present, 0 if empty.

		GT θ_j		
		θ_1	θ_2	$\emptyset$
pred $\hat{\theta}_i$	$\hat{\theta}_1$	0.9	0.2	0.0
	$\hat{\theta}_2$	0.4	0.8	0.0
	$\hat{\theta}_3$	0.6	0.5	0.0

cost $C_{ij} = w_j \|\hat{\theta}_i - \theta_j\|^2$

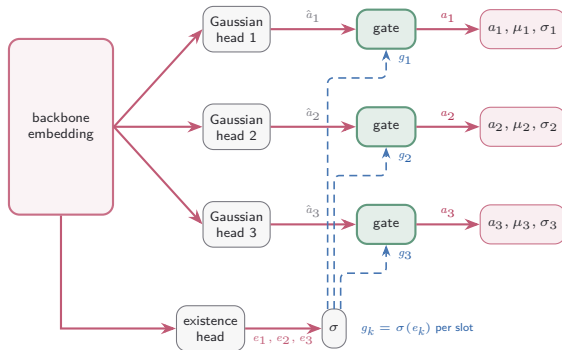
$\xrightarrow[\text{min cost}]{\pi^*}$

		θ_1	θ_2	$\emptyset$
pred $\hat{\theta}_i$	$\hat{\theta}_1$	0.9	0.2	0.0
	$\hat{\theta}_2$	0.4	0.8	0.0
	$\hat{\theta}_3$	0.6	0.5	0.0

$\sum C_{\pi^*} = 0.6$

Column $\emptyset$ is a placeholder GT ($w_j=0$) — all-zero, so the leftover prediction matches it for free: $\hat{\theta}_1 \rightarrow \theta_2$, $\hat{\theta}_2 \rightarrow \theta_1$, $\hat{\theta}_3 \rightarrow \emptyset$.

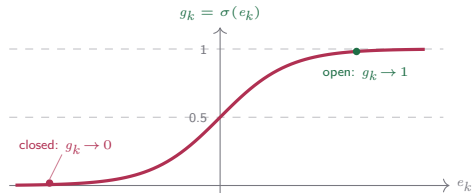
Inside the set-prediction head — gate, blend, match



each gate: $a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$
 μ_k, σ_k pass through $\cdot a_{\text{off}}$ learned per slot

The gate is a sigmoid on the existence logit e_k :

- Open slots ($g_k \rightarrow 1$): $a_k \approx \hat{a}_k$ — the raw regression survives.
- Closed slot ($g_k \rightarrow 0$): $a_k \rightarrow a_{\text{off},k} \approx 0$ whatever $\hat{a}_k$ is, collapsing onto the empty-set target at near-zero loss.
- e_k is internal — the head still emits $3K$ channels (a_k, μ_k, σ_k); a_{off} is a learned per-slot scalar (init 0).



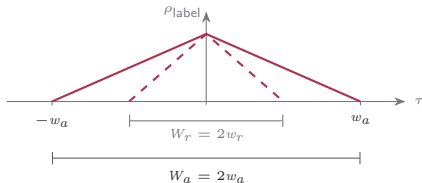
Reading the reach — the network estimates statistics

two labels at offset Δ split into shared and private looks:

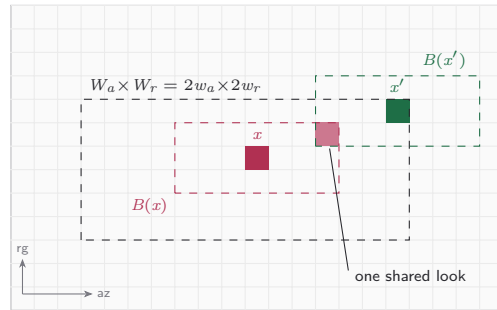
$$w_a w_r \hat{\mathbf{R}}(x) = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x}$$

$$w_a w_r \hat{\mathbf{R}}(x') = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x'}$$

the shared fraction at every offset τ is the correlation law:



the label correlates over $\pm w_a$ in azimuth and $\pm w_r$ in range
— the centred patch spanning both triangles is $W_a \times W_r$.



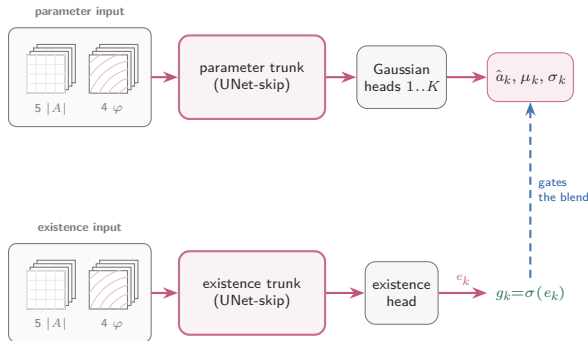
x' sits at the far corner of the patch: $B(x')$ keeps a single shared look with $B(x)$ — one step further and the windows are disjoint ($|\Delta_a| = w_a$ or $|\Delta_r| = w_r$).

The board at five seeds — the L1 family owns the top nine

#	model	loss	curve $R^2 \uparrow$	cosine $\uparrow$	peak err $\downarrow$	recall $\uparrow$	score
1	unet-skip	L1-curve	0.756 $\pm .012$	0.924 $\pm .007$	1.29 $\pm .05$	0.886 $\pm .006$	0.983
2	unet-skip	Charbonnier	0.751 $\pm .012$	0.928 $\pm .004$	1.25 $\pm .03$	0.886 $\pm .004$	0.977
3	unet-skip	param-L1	0.741 $\pm .016$	0.931 $\pm .001$	1.38 $\pm .03$	0.893 $\pm .004$	0.963
4	resunet	param-L1	0.734 $\pm .013$	0.927 $\pm .007$	1.32 $\pm .07$	0.894 $\pm .003$	0.946
5	deeplab	param-L1	0.731 $\pm .006$	0.923 $\pm .004$	1.35 $\pm .05$	0.888 $\pm .002$	0.937
6	deeplab	L1-curve	0.731 $\pm .012$	0.921 $\pm .013$	1.33 $\pm .07$	0.873 $\pm .022$	0.935
7	resunet	L1-curve	0.737 $\pm .019$	0.916 $\pm .007$	1.34 $\pm .06$	0.881 $\pm .005$	0.930
8	resunet	Charbonnier	0.731 $\pm .016$	0.921 $\pm .014$	1.31 $\pm .08$	0.881 $\pm .006$	0.929
9	deeplab	Charbonnier	0.735 $\pm .008$	0.913 $\pm .015$	1.62 $\pm .45$	0.868 $\pm .037$	0.912
10	nafnet	Charbonnier	0.702 $\pm .009$	0.906 $\pm .007$	1.48 $\pm .05$	0.875 $\pm .008$	0.829
11	segformer	Charbonnier	0.688 $\pm .022$	0.910 $\pm .016$	1.51 $\pm .20$	0.866 $\pm .019$	0.829
12	nafnet	L1-curve	0.698 $\pm .007$	0.908 $\pm .010$	1.43 $\pm .08$	0.879 $\pm .005$	0.828
⋮							
16	deeplab	MSE-curve	0.732 $\pm .010$	0.872 $\pm .019$	1.62 $\pm .08$	0.869 $\pm .004$	0.785
20	unet	L1-curve	0.660 $\pm .015$	0.897 $\pm .026$	1.51 $\pm .28$	0.870 $\pm .029$	0.760
21	unet-skip	param-MSE	0.647 $\pm .047$	0.879 $\pm .017$	1.59 $\pm .13$	0.881 $\pm .022$	0.692

runs: model · set-pred · Hungarian · $K=2$ · hv flips · presence A · loss · patch 64×32 , five seeds; held-out band. Cells: mean $\pm$ seed std; **bold** = best mean in column (rows shown), marked only where best and worst differ by more than 5%. score = equal-weight mean of the five metric groups (curve error, variance, SSIM, cosine, peak), min-max scaled over the 35 arms; # = rank; ranks 13–15, 17–19, 22–35 on the next frame only. cosine = per-pixel profile cosine mean; peak err in elevation units; recall = matched. **green** = the winner's score; **red** = the param-MSE reference cell, 21st of 35.

The dual model — two trunks, one head



$$\text{each gate: } a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$$

the set-prediction head of Act III, unchanged — only the source of each embedding splits

- **Split the encoder, keep the head:** two independent UNet-skip trunks — the Gaussian heads read one embedding, the existence head the other. Gate blend and Hungarian matching stay as they are.
- **Each trunk picks its own stack:** the channel groups $|A|$ (5 amplitudes) and φ (4 interferograms) are selected per trunk — which modality reaches which head becomes an architectural choice.
- **Why route:** the input ablation split the labour — phase fires the slots, the full stack calibrates the parameters. The dual model can hand each head the input its job needs.

Pairing the objective — param-L1 plus a curve term

$$\mathcal{L}(w) = (\mathcal{L}_{\text{param-L1}} + w \mathcal{L}_{\text{curve}}) / (1 + w)$$



1 baseline + 15 pair arms per K , five seeds each, at $K=2$ and $K=5$ = 160 runs

The ratio block left the $K=5$ gate under-firing at every split — more existence capacity did not wake it. This experiment changes the objective instead: keep param-L1 and add a synthesized-curve term.

- **Two signals, one loss:** param-L1 reaches a slot only through its matched pair; the curve term re-synthesizes the profile from *all* slots and is permutation-invariant — its gradient touches every slot, matched or not.
- **Magnitude or shape:** L1 and Charbonnier price the synthesized profile's magnitude; cosine normalizes both curves first and prices shape alone.
- **A normalized pair:** the two terms are averaged, so the validation loss changes meaning with w — arms compare on held-out metrics only, never on the loss.
- **The question:** if the conservative gate is starved of a profile-level error signal rather than capacity, a small w should move detection at $K=5$ and leave $K=2$ alone.

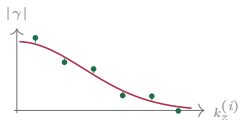
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

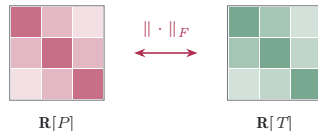
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

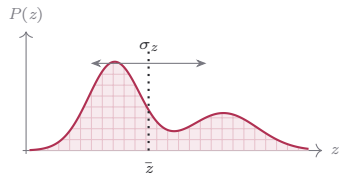
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n) \Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



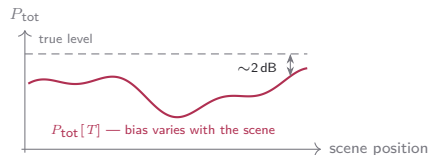
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta\xi$$

$$\ell_{\text{pow}} = \left\langle (P_{\text{tot}}[P] - P_{\text{tot}}[T])^2 \right\rangle$$

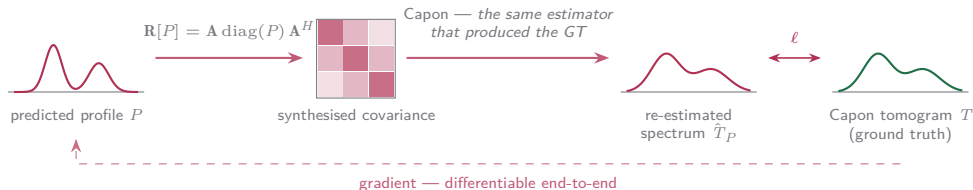


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

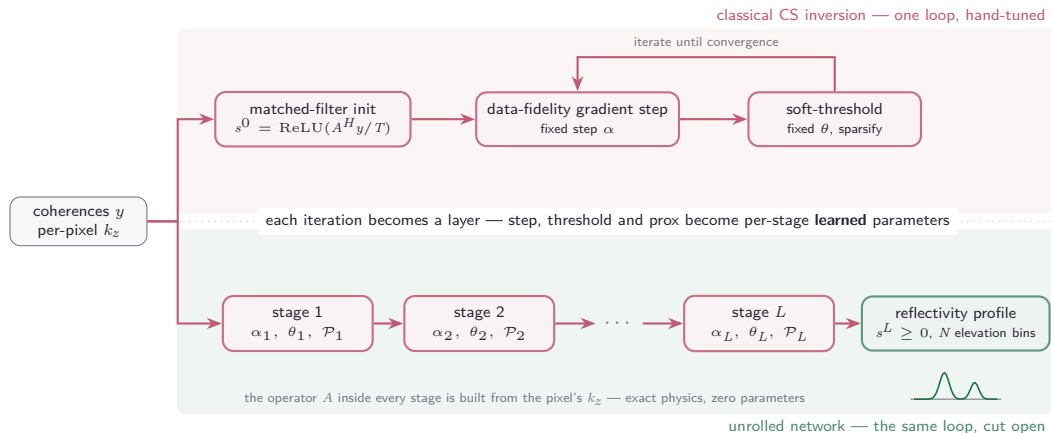
$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

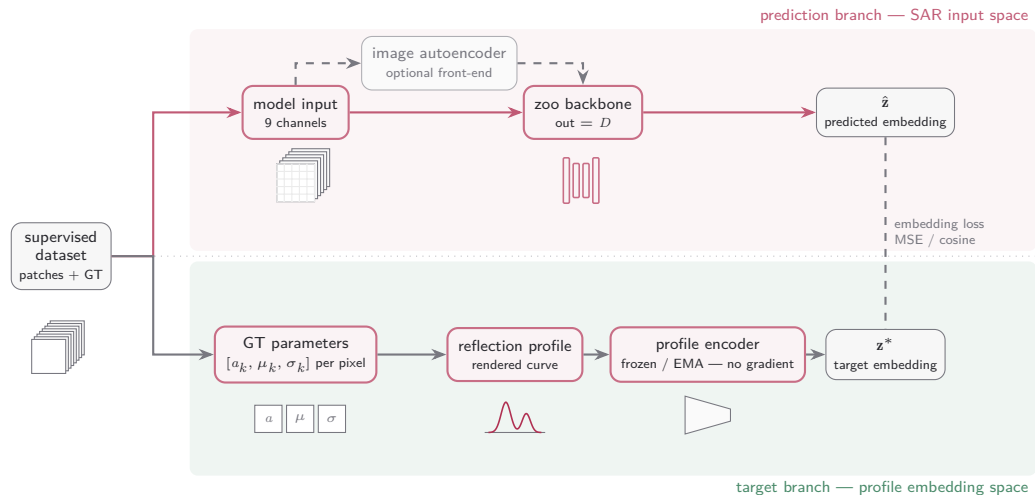
The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \text{Re}[A^H(y - As^\ell)] / L_{\text{lip}}, \quad s^{\ell+1} = \text{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

γ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\text{lip}} = TN \text{d}z^2$ normalises the step. Default $L=8$: $\sim 1.4\text{k}$ parameters.

What JEPA does



A network learns to predict, from the reduced stack alone, the **embedding** that a separately-trained profile autoencoder assigns to the ground-truth reflection profile.